Sang Hoon Lee (이상훈)

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Google Scholar Profile:
http://scholar.google.com/citations?user=GuruM50AAAAJ

Personal Data

Date and Place of Birth: December 21, 1982, Ulsan, Republic of Korea

Nationality: Republic of Korea

Language: English (fluent in writing and speaking), Korean (native language)

Professional Experience

Associate Professor, Department of Physics, Gyeongsang National University (GNU), March 2022–present.

Assistant Professor, Department of Physics, GNU, September 2021–February 2022.

Assistant Professor, Department of Liberal Arts, GNU, March 2021–August 2021.

Assistant Professor, Department of Liberal Arts, Gyeongnam National University of Science and Technology (GNTECH), March 2018–February 2021.

Research Fellow, School of Physics, KIAS, August 2015–February 2018.

BK21+ Research Professor, Department of Energy Science, Sungkyunkwan University (SKKU), August 2014–July 2015.

Postdoctoral Research Associate, Mathematical Institute, University of Oxford, June 2012–June 2014.

Postdoctoral Researcher, Integrated Science Lab (IceLab), Department of Physics, Umeå University, July 2010–June 2012.

Postdoctoral Researcher, Department of Physics, Korea Advanced Institute of Science and Technology (KAIST), February 2010–June 2010.

Education

Ph.D. Physics, KAIST, January 2010.

Dissertation: A few applications and critical phenomena on networks.

Ph.D. Supervisor: Hawoong Jeong (정하웅)

Committee: Hawoong Jeong (chair), Hie-Tae Moon (문희태), Tae-Young Yoon (윤태영), Jae Dong Noh (노재동), and Hyunsuk Hong (홍현숙).

B.S. Physics (Minor: Mathematics), *Summa Cum Laude*, Class Valedictorian, KAIST, February 2004.

Selected Honors and Awards

- Yong-Bong prize (용봉상: 龍峰賞), the Korean Physical Society, 2020.
- Full scholarship at International Summer School: Fundamental Problems in Statistical Physics XII (FPSPXII), Leuven, Belgium, August–September 2009.

- Grand prize, the 1st Korean BioInformation Center (KOBIC) Bioinformatics Contents Contest, February 2007.
- Best oral presentation award, the Korean Physical Society, October 2006.
- Outstanding Teaching Assistant award, KAIST, September 2006.
- Scholarship from Kim Bojeong Basic Science Foundation, September 2004–December 2005.
- Outstanding Intellectuals of the 21st Century Awards, Ministry of Education (Korea), February 2004.
- Seokrim Scholarship from KAIST Alumni, December 2003.

Research Grants as the Principal Investigator

- "Investigating higher-order networks for the machine-learning performance optimization and the application to neural networks", The Basic Science Research Program (핵심연구A) funded by the National Research Foundation of Korea, March 2026–February 2031 (RS-2026-25468383).
- "Renormalization group of networks endowed with potential", Basic Research Laboratory (기초연구실), joint with YongSeok Jho (조용석) and Jaeyoon Cho (조재윤), funded by the National Research Foundation of Korea, June 2022–February 2025 (RS-2022-NR070746).
- "Investigation on mesoscale properties of networks and its application to machine learning", The Basic Science Research Program (우수신진연구) funded by the National Research Foundation of Korea, March 2021–February 2026 (RS-2021-NR061247).
- "Information theoretical approaches on complex networks", The Basic Science Research Program (생애 첫 연구) funded by the National Research Foundation of Korea, September 2018–August 2021 (RS-2018-NR032109).
- "Statistical physics approaches to complex and nonequilibrium systems", Start-up Research Grant funded by GNTECH, March 2018–February 2019.

Research Area

network science; complex systems; statistical physics.

Research Papers

61. J. Lee, D. Lee, S.-W. Son, **S. H. Lee**, M. J. Lee, and H. Kim, "Energy-barrier characterization of secured basins in coupled oscillators with inertia", e-print arXiv:2610.11147, submitted to Chaos Soliton Fract. (under review)
60. Y. Arya and **S. H. Lee**, "Effects of relational graph modularity and depth on the learning performance of neural networks", e-print arXiv:2507.10005, J. Korean Phys. Soc. doi:10.1007/s40042-026-01730-5 (the online-first version).
59. M.-K. Chae, W.-S. Son, and **S. H. Lee**, "Evaluating the impact of incomplete contact tracing data on urban epidemic dynamics", Sci. Rep. doi:10.1038/s41598-026-66867-6 (the early-access version).
58. H. S. Lee, D.-S. Lee, **S. H. Lee**, S. Suweis, and H. J. Park, "Ecological networks of viable species with degree-dependent interaction", Phys. Rev. E **114**, 054307 (2026).
57. **S. H. Lee**, "Friendship-paradox paradox: Do most people's friends really have more friends than they do?", J. Korean Phys. Soc. **88**, 890 (2026).
56. **S. H. Lee**, "Two variants of the friendship paradox: The condition for inequality between them", New Phys.: Sae Mulli **76**, 169 (2026).

55. G. Gim, J. Yun, and **S. H. Lee**, "Quantifying interdisciplinary synergy in higher STEM education", EPJ Data Sci. **14**, 56 (2025).

54. W. Jeong, U. Yu, and **S. H. Lee**, "Global decomposition of networks into multiple cores formed by local hubs", Phys. Rev. E **111**, 054302 (2025).

53. W. Jeong, D. Lee, H. Kim, and **S. H. Lee**, "Kaleidoscopic reorganization of network communities across different scales", Phys. Rev. E **111**, 014312 (2025).

52. S. Jung, **S. H. Lee**, and J. Cho, "Renormalization of complex networks with partition functions", Phys. Rev. E **110**, 064316 (2024).

51. J. I. Park, D.-S. Lee, **S. H. Lee**, and H. J. Park, "Incorporating heterogeneous interactions for ecological biodiversity", Phys. Rev. Lett. **133**, 198402 (2024).

50. G. Gim and **S. H. Lee**, "Statistical analysis on scale and regional distribution of undergraduate physics programs in Korean universities (대한민국 대학의 물리학 학부과정 학과 및 학생 수의 증감과 지역적 분포 분석)", New Phys.: Sae Mulli **74**, 610 (2024).

49. S. Son, E. J. Choi, and **S. H. Lee**, "Revisiting small-world network models: Exploring technical realizations and the equivalence of the Newman–Watts and Harary models", J. Korean Phys. Soc. **83**, 879 (2023).

48. J. Seo, G. Gim, S. Kim, M. Ha, and **S. H. Lee**, "Scaling relations between regional school-age population density and densities of college departments and students by majors (지역별 학령인구밀도와 대학 전공별 학과 및 학생 밀도의 스케일링 관계)", New Phys.: Sae Mulli **73**, 858 (2023).

47. D. Bernenko, **S. H. Lee**, and L. Lizana, "Exploring 3D community inconsistency in human chromosome contact networks", J. Phys.: Complex. **4**, 035004 (2023).

46. D. Bernenko, **S. H. Lee**, P. Stenberg, and L. Lizana, "Mapping the semi-nested community structure of 3D chromosome contact networks", PLOS Comput. Biol. **19**, e1011185 (2023).

45. M. J. Lee, J.-H. Kim, K.-I. Goh, **S. H. Lee**, S.-W. Son, and D.-S. Lee, "Degree distributions under general node removal: Power-law or Poisson?", Phys. Rev. E **106**, 064309 (2022).

44. **S. H. Lee**, "Refinement for community structures of bipartite networks", J. Korean Phys. Soc. **79**, 1190 (2021).

43. M. J. Lee, E. Lee, B. Lee, H. Jeong, D.-S. Lee, and **S. H. Lee**, "Uncovering hidden dependency in weighted networks via information entropy", Phys. Rev. Res. **3**, 043136 (2021).

42. D. Lee, **S. H. Lee**, B. J. Kim, and H. Kim, "Consistency landscape of network communities", Phys. Rev. E. **103**, 052306 (2021).

41. A. M. Martín González, D. P. Vázquez, R. Ramos-Jiliberto, **S. H. Lee**, and V. Miele, "Core-periphery structure in mutualistic networks: an epitaph for nestedness?", bioRxiv doi:10.1101/2020.04.02.021691.

40. **S. H. Lee**, "Mesoscale properties of mutualistic networks in ecosystems (생태계 상호 도움 네트워크의 중간 크기 성질 분석)", New Phys.: Sae Mulli **70**, 1077 (2020).

39. J. Lee, **S. H. Lee**, J. Um, and H. Park, "Carnot efficiency and zero-entropy-production rate do not guarantee reversibility of a process", J. Korean Phys. Soc. **75**, 948 (2019).

38. R. Flanagan, L. Lacasa, E. K. Towlson, **S. H. Lee**, and M. A. Porter, "Effect of antipsychotics on community structure in functional brain networks", J. Complex Netw. **7**, 932 (2019).

37. H. Kim, M. J. Lee, **S. H. Lee**, and S.-W. Son, "On structural and dynamical factors determining the integrated basin instability of power-grid nodes", Chaos **29**, 103132 (2019).

36. H. Kim and **S. H. Lee**, "Relational flexibility of network elements based on inconsistent community detection", Phys. Rev. E **100**, 022311 (2019).

35. **S. H. Lee**, Y. Kim, S. Lee, X. Durang, J.-H. Jeon, P. Stenberg, and L. Lizana, "Mapping the spectrum of 3D communities in human chromosome conformation capture data", Sci. Rep. **9**, 6859 (2019).

34. J. Yun, **S. H. Lee**, and H. Jeong, "Early onset of structural inequality in the formation of collaborative knowledge in all Wikimedia projects", Nat. Hum. Behav. **3**, 155 (2019).

33. **S. H. Lee** and P. Holme, "Navigating temporal networks", Physica A **513**, 288 (2019).

32. **S. H. Lee**, J. Um, and H. Park, "Nonuniversality of heat-engine efficiency at maximum power", Phys. Rev. E **98**, 052137 (2018).

31. H. Kim, **S. H. Lee**, J. Davidsen, and S.-W. Son, "Multistability and variations in basin of attraction of power-grid systems", New J. Phys. **20**, 113006 (2018).

30. H. J. Park, W. S. Jo, **S. H. Lee**, and B. J. Kim, "Generalized gravity model for human migration", New J. Phys. **20**, 093018 (2018).

29. X.-M. Cui, C. N. Yoon, H. Youn, **S. H. Lee**, J. S. Jung, and S. K. Han, "Dynamic burstiness of word-occurrence and network modularity in textbook systems", Physica A **487**, 103 (2017).

28. E. Lee, P. Holme, and **S. H. Lee**, "Modeling the dynamics of dissent", Physica A **486**, 262 (2017).

27. **S. H. Lee**, M. D. Fricker, and M. A. Porter, "Mesoscale analyses of fungal networks as an approach for quantifying phenotypic traits", J. Complex Netw. **5**, 145 (2017).

26. **S. H. Lee**, J. M. Magallanes, and M. A. Porter, "Time-dependent community structure in legislation cosponsorship networks in the Congress of the Republic of Peru", J. Complex Netw. **5**, 127 (2017).

25. **S. H. Lee**, J. Um, and H. Park, "Efficiency at the maximum power output for simple two-level heat engine", e-print arXiv:1612.00518.

24. M. Cucuringu, P. Rombach, **S. H. Lee**, and M. A. Porter, "Detection of core-periphery structure in networks using spectral methods and geodesic paths", Eur. J. Appl. Math. **27**, 846 (2016).

23. H. Kim, **S. H. Lee**, and P. Holme, "Building blocks of basin stability of power grids", Phys. Rev. E **93**, 062318 (2016).

22. **S. H. Lee**, "Network nestedness as generalized core-periphery structures", Phys. Rev. E **93**, 022306 (2016).

21. J. Yun, **S. H. Lee**, and H. Jeong, "Intellectual interchanges in the history of the massive online open-editing encyclopedia, Wikipedia", Phys. Rev. E **93**, 012307 (2016).

20. H. Kim, **S. H. Lee**, and P. Holme, "Community consistency determines the stability transition window of power-grid nodes", New J. Phys. **17**, 113005 (2015).

19. **S. H. Lee**, R. Ffrancon, D. M. Abrams, B. J. Kim, and M. A. Porter, "Matchmaker, matchmaker, make me a match: Migration of populations via marriages in the past", Phys. Rev. X **4**, 041009 (2014).

18. **S. H. Lee**, M. Cucuringu, and M. A. Porter, "Density-based and transport-based core-periphery structures in networks", Phys. Rev. E **89**, 032810 (2014).

17. J. Zhao, **S. H. Lee**, M. Huss, and P. Holme, "The network organization of cancer-associated protein complexes in human tissues", Sci. Rep. **3**, 1583 (2013).

16. **S. H. Lee** and P. Holme, "A greedy-navigator approach to navigable city plans", Eur. Phys. J.-Spec. Top. **215**, 135 (2013).

15. **S. H. Lee** and P. Holme, "Geometric properties of graph layouts optimized for greedy navigation", Phys. Rev. E **86**, 067103 (2012).

14. **S. H. Lee** and P. Holme, "Exploring maps with greedy navigators", Phys. Rev. Lett. **108**, 128701 (2012) [selected as Editors' Suggestions].

13. **S. H. Lee**, S. Bernhardsson, P. Holme, B. J. Kim, and P. Minnhagen, "Neutral theory of chemical reaction networks", New J. Phys. **14**, 033032 (2012) [selected as IOPselect & featured in the Highlights of 2012 collection of New J. Phys.].
12. **S. H. Lee**, S. Lee, S.-W. Son, and P. Holme, "Phase-shift inversion in oscillator systems with periodically switching couplings", Phys. Rev. E **85**, 027202 (2012).
11. **S. H. Lee** and P. Holme, "Pathlength scaling in graphs with incomplete navigational information", Physica A **390**, 3996 (2011).
10. **S. H. Lee**, P.-J. Kim, and H. Jeong, "Global organization of protein complexome in the yeast *Saccharomyces cerevisiae*", BMC Syst. Biol. **5**, 126 (2011) [designated as highly accessed articles].
9. P. Holme, M. Huss, and **S. H. Lee**, "Atmospheric reaction systems as null-models to identify structural traces of evolution in metabolism", PLOS ONE **6**(5), e19759 (2011).
8. **S. H. Lee**, M. Ha, C. Jeon, and H. Jeong, "Finite-size scaling in random *K*-satisfiability problems", Phys. Rev. E **82**, 061109 (2010).
7. Y. Baek, **S. H. Lee**, and H. Jeong, "Market behavior and performance of different strategy evaluation schemes", Phys. Rev. E **82**, 026109 (2010).
6. **S. H. Lee**, P.-J. Kim, Y.-Y. Ahn, and H. Jeong, "Googling social interactions: Web search engine based social network construction", PLOS ONE **5**(7), e11233 (2010).
5. **S. H. Lee**, M. Ha, H. Jeong, J. D. Noh, and H. Park, "Critical behavior of the Ising model in annealed scale-free networks", Phys. Rev. E **80**, 051127 (2009).
4. **S. H. Lee**, H. Jeong, and J. D. Noh, "Random field Ising model on networks with inhomogeneous connections", Phys. Rev. E **74**, 031118 (2006).
3. **S. H. Lee** and H. Jeong, "Effects of substrate network topologies on competition dynamics", Phys. Rev. E **74**, 026118 (2006).
2. **S. H. Lee** and H. Jeong, "Minority game with interaction via various networks", J. Korean Phys. Soc. **48**, S186 (2006).
1. **S. H. Lee**, P.-J. Kim, and H. Jeong, "Statistical properties of sampled networks", Phys. Rev. E **73**, 016102 (2006).

Translation

- 복잡계 네트워크의 자연법칙 (에이콘출판사, 2026), in collaboration with M. J. Lee (이미진), H. Kim (김희태), S.-W. Son (손승우), J. Yun (윤진혁), and E. Lee (이은).
  - the original book: S. N. Dorogovtsev and J. F. F. Mendes, *The Nature of Complex Networks* (Oxford University Press, 2022).
  - selected as "Excellent Academic Book (우수학술도서)" by National Academy of Sciences of the Republic of Korea (대한민국학술원) in 2026.
- 비선형 동역학과 카오스 2/e (에이콘출판사, 2025), in collaboration with H. J. Park (박혜진), H. Kim (김희태), M. J. Lee (이미진), and S.-W. Son (손승우).
  - the original book: Steven H. Strogatz, *Nonlinear Dynamics and Chaos: 2nd Edition* (Westview Press, 2014).
- 네트워크 사이언스 (에이콘출판사, 2023), in collaboration with M. J. Lee (이미진), J. Yun (윤진혁), E. Lee (이은), H.-H. Jo (조항현), H. Kim (김희태), and S.-W. Son (손승우).

  - the original book: A.-L. Barabási, *Network Science* (Cambridge University Press, 2016).
  - selected as "Excellent Academic Book (우수학술도서)" by National Academy of Sciences of the Republic of Korea (대한민국학술원) in 2024.
- 네트워크 2/e (에이콘출판사, 2022), in collaboration with H. Kim (김희태), S.-W. Son (손승우), J. Yun (윤진혁), H.-H. Jo (조항현), E. Lee (이은), and M. J. Lee (이미진).
  - the original book: M. E. J. Newman, *Networks: 2nd Edition* (Oxford University Press, 2018).
- 네트워크 분석 (에이콘출판사, 2022), in collaboration with S.-W. Son (손승우), Y.-H. Eom (엄영호), E. Lee (이은), H. Kim (김희태), and M. J. Lee (이미진).
  - the original book: F. Menczer, S. Fortunato, and C. A. Davis, *A First Course in Network Science* (Cambridge University Press, 2020).
  - selected as "Excellent Academic Book (우수학술도서)" by National Academy of Sciences of the Republic of Korea (대한민국학술원) in 2022.
- The Korean version of Network Literacy Project: 네트워크 이해하기, in collaboration with M. J. Lee (이미진).

Media Coverage

- "Systematic investigation on the Wikimedia projects reveals the structural inequality in collaborative knowledge formation", KAIST Compass, April 23, 2019.
- "Inequalities in crowdsourced knowledge", Editorial in Nat. Hum. Behav., February 12, 2019. `https://www.nature.com/articles/s41562-019-0550-5`.
- "Study Shows a Monopolizing Few Can Control Collective Intelligence", The Korea Bizwire, December 20, 2018. `http://koreabizwire.com/study-shows-a-monopolizing-few-can-control-collective-intelligence/129098`
- "How I met my collaborators", Behavioural & Social Sciences at Nature Research, December 17, 2018. `https://socialsciences.nature.com/users/199227-jinhyuk-yun/posts/42267-how-i-met-my-collaborators`
- "Wikipedia's social structures resemble a bureaucratic corporation, studies say", The Washington Post, April 28, 2016. `http://wapo.st/1Sx0UQG`
- "Wikipedia has become 20th century bureaucracy", Fudzilla, April 28, 2016. `http://www.fudzilla.com/news/40552-wikipedia-has-become-20th-century-bureaucracy`
- "Wikipedia is basically just another giant bureaucracy, study finds", ScienceAlert, April 28, 2016. `http://www.sciencealert.com/wikipedia-is-basically-just-another-old-fashioned-bureaucracy-study-finds`
- "Wikipedia is basically a corporate bureaucracy, according to a New study", Gizmodo, April 25, 2016. `http://gizmodo.com/wikipedia-is-basically-a-corporate-bureaucracy-accordi-1746955234`
- "What the editing history of Wikipedia reveals", Asian Scientist, February 15, 2016. `http://www.asianscientist.com/2016/02/in-the-lab/editing-history-wikipedia-reveals/`
- "Wikipedia vulnerabilities explored in new research", The Speaker, February 4, 2016. `http://thespeaker.co/science/wikipedia-vulnerabilities-explored-new-research/`
- "Wikipedia articles separate into four categories", APS Physics: Focus, January 22, 2016. `http://physics.aps.org/articles/v9/8`
- "Marriage records reveal patterns of Korean migration through the centuries", Pacific Standard, November 4, 2014. `http://www.psmag.com/health-and-behavior/marriage-records-probe-korean-migration-centuries-93632`

- "How wedding registries reveal migration paths", Mathematical Institute website, October 17, 2014. http://www.maths.ox.ac.uk/node/13171
- "Wedding registries reveal migration paths", APS Physics: Synopsis, October 16, 2014. http://physics.aps.org/synopsis-for/10.1103/PhysRevX.4.041009 (introduced in Physics Today facebook page: https://www.facebook.com/PhysicsToday/posts/10154776052125164).
- "Bugs Bunny as mathematician", Mathematical Institute website, September 2, 2014. https://www.maths.ox.ac.uk/node/894
- "Greed is good", APS Physics: Synopsis, March 22, 2012. http://physics.aps.org/synopsis-for/10.1103/PhysRevLett.108.128701
- "Google tracks political allegiances", New Scientist: issue 2771, p19, July 31, 2010. http://www.newscientist.com/article/mg20727715.200
- "Pocket Science Ep1 – Google something new", http://www.youtube.com/watch?v=Mpp_kZrA-vA (YouTube), July 26, 2010.

Research Supervision

Postdoc in Research Institute of Natural Science, GNU.

- Wonhee Jeong (정원희), September 2023–January 2025.

Supervising graduate students in Department of Physics, GNU.

- Seungho Choi (최승호), master's course, September 2026–present.
- Daon Kim (김다온), master's course, March 2026–present.
- Gahyoun Gim (김가현), master of science in physics, March 2024–February 2026.
- Yash Arya, master of science in physics, September 2023–August 2025.
- Seohyeon Kim (김서현), master's course, September 2022–June 2024.

Supervising undergraduate students in Department of Physics, GNU.

- Seongjun Woo (우승준): Wikipedia in the era of LLM, March 2026–present.
- Jihyeon Jang (장지현): drug–side-effect–food networks, September 2025–present.
- 김민재, 김영규, 이대호: ecological networks and international car-trade networks, September 2025–June 2026.
- Seungho Choi (최승호): soccer network analysis, December 2024–August 2026.
- Daon Kim (김다온): traffic flow in highway networks, September 2024–February 2026.
- Karam Ko (고가람): network analysis to explore the knowledge structure of Physics on Wikipedia, September 2023–August 2024.
- Seora Son (손서라), Eun Ji Choi (최은지): the effect of neural network structures on their learning performance, September 2022–August 2023.
- Gahyoun Gim (김가현): recent temporal changes in curriculum at physics or related departments in Korean universities, March 2022–February 2024.
- Eun Ji Kim (김은지), Ga-Hyeon Lee (이가현), Hye Jeong Song (송혜정): the network analysis of four major rivers in Korea, September 2021–December 2022.

Co-supervising Ph.D. students in Department of Energy Science, Sungkyunkwan University.

- Heetae Kim (김희태), Minjin Lee (이민진), Eun Lee (이은): joint with P. Holme, Summer 2014–Spring 2015.

Co-supervising undergraduate/master's students in Mathematical Institute, University of Oxford.

- Ryan Flanagan (master's student in Mathematical Institute), "Network analysis of separated cognitive states in the human brain", joint with M. A. Porter, Summer 2013–Spring 2014.
- Fabian Ying (undergraduate student in Mathematical Institute), "Dynamical processes on random geometric graphs", joint with M. A. Porter, Summer 2013–Fall 2013.
- Melissa Lever (master's student in Systems Biology Doctoral Training Centre), "Characterising patients and controls with brain graphs constructed from fMRI data", joint with M. A. Porter, Summer 2012.

Assessor of the mini-projects, MSc in Mathematics and the Foundations of Computer Science (MFoCS), University of Oxford.

- Trinity Term (April 21–June 15), 2013.

Examiner (supervising master's thesis), Umeå University.

- Johan Dahlin, "Community detection in imperfect network", Master's Thesis in Engineering Physics, January–June 2011.

Talks and Seminars

Invited Talks and Lectures

- "When does network structure help a neural network learn? Communities, depth, and group interactions", Korea-Japan Scientific Computing Symposium 2026, Pukyong National University, Busan, Korea, December 17, 2026.
- "Harnessing inconsistency: Revealing network intricacies in communities and geometric embeddings", Econosociophysics Research Society Workshop (경제사회물리연구회 워크샵), Pohang University of Science and Technology (POSTECH), Pohang, Korea, November 13, 2026.
- "Harnessing inconsistency: Revealing network intricacies in communities and geometric embeddings", CCP 2026: XXXVII IUPAP Conference on Computational Physics, Soongsil University, Seoul, Korea, August 12, 2026.
- "Network science: from the small-world into the real and artificial worlds", Department of Computer Engineering, Ulsan National Institute of Science and Technology (UNIST), Ulsan, Korea, November 19, 2025.
- "Harnessing inconsistency: Revealing network intricacies in communities and geometric embedding", APCTP/G-LAMP Workshop on the Roles of Heterogeneity in Nonequilibrium (RHINO) 2025, Pukyong National University, Busan, Korea, August 30, 2025.
- "Kaleidoscopic reorganization of network communities across different scales", KIAS-KU International Workshop: Theoretical Challenges in Network Science, Seoul, Korea, November 7, 2024.
- "[Social Physics] Population density vs facility density: the scaling analysis", the Busan-Ulsan-Gyeongnam (BUG) branch conference of the Korean Physical Society 2004, GNU, Jinju, Korea, June 28, 2024.
- "Network science: from the small-world into the real world", The 21st KIAS-APCTP Winter School On Statistical Physics, POSTECH, Pohang, Korea, January 8–12, 2024.
- "Information theoretical perspectives on networks: mutuality and complexity", STATPHYS28 Satellite Meeting: New Frontiers in Complex Networks, Pukyong National University, Busan, Korea, August 4, 2023.
- "Scale-dependent landscape of semi-nested community structures of 3D chromosome contact networks", PhysNet23: the 1st Workshop on Physical Networks (a satellite meeting of NetSci 2023: International School and Conference on Network Science), Vienna, Austria, July 11, 2023.
- "Mesoscale properties of mutualistic networks in ecosystems", Modeling on Ecology and Evolution Workshop (MEEW), Seoul, Korea, October 28, 2022.

- "Scale-dependent landscape of semi-nested community structures of 3D chromosome contact networks", KIAS-KU International Workshop: Theoretical Challenges in Network Science, Seoul, Korea, September 29, 2022.
- "…but is this physics? Physicists' journey through digital or digitized data", Soongsil University (online), Seoul, Korea, May 25, 2022. https://youtu.be/LhXueVuic4Q (in Korean)
- "The World through the Research on Statistcal Physics and Complex Networks (통계물리와 네트워크 연구로 보는 세상)", Chosun University, Gwangju, Korea, April 1, 2022.
- "Community structures of chromosome conformation networks across various scales", Frontiers in Theoretical Biophysics, online/Gyeongju, Korea, November 18, 2021. https://youtu.be/kMxrNPzyBoc
- "Mining stochasticity of community detection to refine global and local organization of networks", Non-equilibrium collective phenomena workshop, online/Gyeongju, Korea, September 29, 2021. https://youtu.be/omXuRKxH7iA
- "Mining stochasticity of community detection to refine global and local organization of networks", Roles of heterogeneity in non-equilibrium collective dynamics (RHINO 2021), online/APCTP, Korea, August 17, 2021. https://youtu.be/NYwx3O17yjU
- "Mining stochasticity of community detection to refine global and local organization of networks", StatPhys Monthly Meeting, online, July 19, 2021. https://youtu.be/06oJE55v-2c (in Korean)
- "Mesoscale properties of mutualistic networks in ecosystems", The 2nd workshop on stochasticity and fluctuations in small systems, online, November 26, 2020. https://youtu.be/Z4dDhswtixY
- "Structural properties of ecological networks and their effect on dynamical stability", 2020 APCTP Modeling on Ecology and Evolution Workshop, online, August 25, 2020. https://youtu.be/1sz3GF_3bgs (in Korean)
- "Structural inequality and polarization in collaborative knowledge and opinion formation", Roles of heterogeneity in non-equilibrium collective dynamics (RHINO 2020), online, August 19, 2020. https://youtu.be/z7kLVl1jbbY
- "Network science: Standing on the edges of a giant collaboration network", 2020 The Korean Physical Society Spring Meeting, online, July 14, 2020. https://youtu.be/mnZjSZ8EfAQ (in Korean)
- "Ecological networks: Nested or not—or does it really matter?", 2020 APCTP Seminar Series: Modeling on Ecology and Evolution (MEE), Asia Pacific Center for Theoretical Physics (APCTP), Pohang, Korea, July 6, 2020. https://youtu.be/MAJEzTsdDBU (in Korean)
- "Uncovering hidden mutual importance in networks via information entropy", KAIST AI+X [Applied Science] Symposium 2019, Institute for Basic Science (IBS), Daejeon, Korea, June 21, 2019. https://youtu.be/1xk6EJ3R3sk
- "Effect of antipsychotics on community structure in functional brain networks", International Workshop on Function, Information Spreading, and Percolation in Brain Networks, APCTP, Pohang, Korea, May 15, 2019.
- "Mapping the spectrum of 3D communities in human chromosome conformation capture data", The Arctic Biophysics Meeting on Epigenetics and Chromosome Dynamics, Kiruna, Sweden, January 22, 2019.
- "Uncovering hidden mutual importance in networks via information entropy", International Workshop on Theoretical Perspectives in Network Science, Seoul National University, Korea, December 8, 2018.
- "Uncovering hidden mutual importance in networks via information entropy", The 6th International Workshop on Physics of Social Complexity (PoSCo), POSTECH, Pohang, Korea, June 1, 2018.
- "Modeling the dynamics of dissent", Perspectives in Statistical Physics on Evolutionary Game Theory, Pukyong National University, Busan, Korea, May 6, 2018.

- "Finding higher-order interactions in human chromosomes", 2018 The Korean Physical Society Spring Meeting, Daejeon, Korea, April 26, 2018.
- "Finding higher-order interactions in human chromosomes", A Mini-Symposium on the Biophysics of Human Chromosome, POSTECH, Pohang, Korea, February 5, 2018.
- "Higher-order network structures in topologically associated domains of chromosome interactions", When Complex Networks Meet Complex Data: Higher-Order Models in Network Science (NetSci2017 Satellite Workshop), Indianapolis, the United States, June 20, 2017.
- "Higher-order organization of interactions in chromosome: 1D sequence → 3D structure → topologically associated domain via network community detection", A mini-symposium on structures and dynamics in human chromosomes, POSTECH, Pohang, Korea, March 27, 2017.
- "Network education in this country: my personal perspective", NetSciEd5: The Fifth NetSci Satellite Symposium on Network Science and Education, Seoul, Korea, May 31, 2016.
- "Networks: From the small world into the real world", Computational Neuroscience Winter School 2015, POSTECH, Pohang, Korea, February 5, 2015.
- "Mesoscale analyses of political networks", Physics of Social Complexity (PoSCo), POSTECH, Pohang, Korea, January 28, 2015.
- "Migration of populations via marriages in the past", The 4th Thermoelectric Workshop (TEW) in Korea Electrotechnology Research Institute (KERI), Gyeongju, Korea, December 18, 2014.
- "Investigation on the marriages and migration in the past based on Korean family books and modern census (족보, 인구조사 자료를 기반으로 한 본관별 혼인과 인구 이동에 대한 연구 in Korean)", The Fall Conference of Korea Academy of Complexity Studies, Ewha Womans University, Seoul, Korea, November 29, 2014.
- "Matchmaker, matchmaker, make me a match: Migration of populations via marriages in the past", The First Wednesday Multidisciplinary Forum, KAIST, Daejeon, Korea, November 5, 2014.
- "Matchmaker, matchmaker, make me a match: Migration of populations via marriages in the past", Invited Session on Social Simulation, the Korean Operations Research and Management Science Society (KORMS) Fall Conference, Korea University, Seoul, Korea, November 1, 2014.
- "Physics of connection: Beyond the nearest neighbor towards the era of friend recommendation (연결의 물리학: 일촌의 파도를 넘어 친구 추천의 시대로 in Korean)", Colloquium at the Subject Matter Education Institute of Chosun University, Gwangju, Korea, October 28, 2014.
- "Exploring road networks with greedy navigators and their core-periphery structures", an invited seminar at the research group on Large Graphs and Networks in Université catholique de Louvain, Louvain-la-Neuve, Belgium, December 14, 2012.
- "Neutral theory of chemical reaction networks", Maximum entropy and ecology: advanced intensive one-week interdisciplinary course at Umeå University, Umeå, Sweden, October 4, 2012. `http://www.org.umu.se/icelab/english/education/maximum-entropy-and-ecology/`.
- "Googling social interactions: Web search engine based social network construction", High Throughput Humanities [satellite meeting at the European Conference on Complex Systems 2010 (ECCS'10)], Lisbon University Institute ISCTE, Lisbon, Portugal, September 15, 2010.
- "Glass transition", The 5th APCTP-KIAS Winter School on Statistical Physics, Phoenix Park, Pyeongchang, Korea, January 30, 2008.

Talks and Poster Presentations (international conferences)

- "Scale-dependent landscape of semi-nested community structures of 3D chromosome contact networks", 28th International Conference on Statistical Physics (STATPHYS 28), The University of Tokyo, Tokyo, Japan, August 9, 2023.
- "Scale-dependent landscape of semi-nested community structures of 3D chromosome contact networks", NetSci 2023: International School and Conference on Network Science, Vienna, Austria, July 12, 2023.
- "Refinement for community structures of bipartite networks", ComNets (Communities in Networks) @ NetSci 2022, online, July 11, 2022. `https://youtu.be/mrYs82QbHTI`
- "Mining stochasticity of community detection to refine global and local organization of networks", Conference on Complex Systems (CCS) 2021, Lyon/online, October 25, 2021. `https://youtu.be/xftNmrKDsVg`
- "Uncovering hidden dependency in weighted networks via information entropy", Networks 2021: A Joint Sunbelt and NetSci Conference, online, July 6, 2021. `https://youtu.be/mZUk6VhWvOg`
- "Uncovering hidden dependency in weighted networks via information entropy", Conference on Complex Systems (CCS) 2020, online, December 8, 2020.
- "Discovering relational flexibility from inconsistent community detection in networks", The 6th International Conference on Computational Social Science (IC^2S^2), online, July 19, 2020. `https://youtu.be/7iZWsdXw-5Y`
- "Uncovering hidden mutual importance in networks via information entropy", NetSciX: International School and Conferences on Network Science, Tokyo, Japan, January 20, 2020.
- "Uncovering hidden mutual importance in networks via information entropy", The 5th International Conference on Computational Social Science (IC^2S^2), Amsterdam, The Netherlands, July 18, 2019.
- "Higher-order interactions in human chromosomes: finding topologically associated domains via network community detection", the 30th Marian Smoluchowski Symposium on Statistical Physics: On the Uniformity of Laws of Nature, Kraków, Poland, September 7, 2017.
- "Nonuniversality of heat engine efficiency at maximum power", the 30th Marian Smoluchowski Symposium on Statistical Physics: On the Uniformity of Laws of Nature, Kraków, Poland, September 5, 2017.
- "Finding topologically associated domains in chromosome interactions via network community identification", NetSci2017: International School and Conference on Network Science, Indianapolis, the United States, June 21, 2017.
- "Efficiency at the maximum power output for simple two-level heat engine", 9th Dynamics Days Asia Pacific (DDAP9), Hong Kong, December 15, 2016.
- "Efficiency at the maximum power output for simple two-level heat engine", Workshop on Stochasticity and Fluctuations in Small Systems, POSTECH, Pohang, Korea, November 30, 2016.
- "Efficiency at the maximum power output for simple two-level heat engine", 26th International Conference on Statistical Physics of the International Union of Pure and Applied Physics (STATPHYS 26), Lyon, France, July 18–22, 2016.
- "Efficiency at the maximum power output for simple two-level heat engine", The 7th KIAS Conference on Statistical Physics, KIAS, Seoul, Korea, July 5, 2016.
- "Core-periphery structures in networks and its relation to network nestedness", NetSci2016: International School and Conference on Network Science, Seoul, Korea, June 2, 2016.
- "Finding Lagrangian coherent structures using community detection", The 3rd East Asia Joint Seminar on Statistical Physics, KIAS, Seoul, Korea, October 15, 2015. video for the talk: `http://media.kias.re.kr/detailPage.do?pro_seq=996&type=p`

- "Matchmaker, matchmaker, make me a match: Migration of populations via marriages in the past", International Conference on Computational Social Science (ICCSS), Helsinki, Finland, June 11, 2015.
- "Finding Lagrangian coherent structures using community detection", NetSciX: International School and Conferences on Network Science, Rio de Janeiro, Brazil, January 15, 2015.
- "Community structures of multilayer political cosponsorship networks", The 2nd Daegu Gyeongbuk (Gyeongju) International Social Network Conference (DISC 2014), Eldis Regent Hotel, Daegu, Korea, December 12, 2014.
- "Matchmaker, matchmaker, make me a match: Migration of populations via marriages in the past", Application of Econophysics and Social Physics: Summer Workshop, Inha University, Incheon, Korea, August 22, 2014.
- "Core-periphery structures in networks based on edge density, transport, and eigenvalue spectra", NetSci2014: International School and Conference on Network Science, Clark Kerr Campus, Berkeley, California, the United States, June 5, 2014.
- "Matchmaker, matchmaker, make me a match: Migration of populations via marriages in the past", TnetSphys'14: Temporal Networks, Human Dynamics and Social Physics (satellite meeting at the NetSci2014: International School and Conference on Network Science), Clark Kerr Campus, Berkeley, California, the United States, June 3, 2014.
- "Density-based and transport-based core-periphery structures in networks", Cambridge Networks Day (CNDay) 2014, the Sainsbury Laboratory, University of Cambridge, Cambridge, United Kingdom, May 23, 2014.
- "Density-based and transport-based core-periphery structures in networks", Dynamics Days US 2014, Georgia Institute of Technology, Atlanta, Georgia, the United States, January 3, 2014.
- "Density and transport-based core-periphery structures in networks", Network Frontier Workshop, Northwestern University, Evanston, Illinois, the United States, December 5, 2013.
- "Overlapping community detection of multilayer networks", Multiplex Networks 2013: Towards the understanding of a complex world [satellite meeting at the European Conference on Complex Systems 2013 (ECCS'13)], World Trade Center, Barcelona, Spain, September 19, 2013.
- "Matchmaker, matchmaker, make me a match: migration of population via marriages in the past", COVEnANT 2013: Collective Behaviors and Networks [satellite meeting at the European Conference on Complex Systems 2013 (ECCS'13)], World Trade Center, Barcelona, Spain, September 19, 2013.
- "Overlapping community detection on multilayer networks", Econophysics Colloquium 2013 & Asia Pacific Econophysics Conference 2013 (satellite meeting of STATPHYS 25), Pohang, Korea, July 30, 2013.
- "Matchmaker, matchmaker, make me a match: migration of population via marriages in the past", 25th International Conference on Statistical Physics of the International Union of Pure and Applied Physics (STATPHYS 25), Seoul, Korea, July 26, 2013.
- "Mesoscopic spatiotemporal properties of growing fungal networks", Uncertainty in Interaction Networks, University of Bath, Bath, United Kingdom, June 13, 2013.
- "The organization of human cancer-related protein complexes", Genomic Complexity [satellite meeting at the European Conference on Complex Systems 2012 (ECCS'12)], Université Libre de Bruxelles, Brussels, Belgium, September 6, 2012.
- "Geographic network in the eyes of greedy path-finding navigators", Nordic Network for Network Science (NNNS) Kickoff, Stockholm, Sweden, November 2011.
- "Exploring spatial networks with greedy navigators", NetSci2011: International School and Conference on Network Science, Central European University, Budapest, Hungary, June 2011.

- "Exploring spatial networks with greedy navigators", Conference on Applications of Network Theory, Nordic Institute for Theoretical Physics (NORDITA), Stockholm, Sweden, April 2011.
- "Exploring spatial networks with greedy navigators", Applications of network theory: from mechanisms to large-scale structure, NORDITA, Stockholm, Sweden, April 2011.
- "Finite-size effects of threshold behavior in the K-satisfiability problem", Dynamics Days 2010: International Conference on Chaos and Nonlinear Dynamics, Northwestern University, Evanston, Illinois, the United States, January 2010.
- "Critical behavior of the Ising model in annealed scale-free networks", Dynamics Days 2010: International Conference on Chaos and Nonlinear Dynamics, Northwestern University, Evanston, Illinois, the United States, January 2010.
- "Finite-size effects of threshold behavior in random K-satisfiability (K-SAT) problem", International Summer School: Fundamental Problems in Statistical Physics XII, La Foresta, Leuven, Belgium, September 2009.
- "Googling social interactions: Web search engine based social network construction", International Conference on Complex Networks: The Past 10 Years and Future, Seoul National University, Seoul, Korea, December 2008.
- "Finite-size effects of phase transitions in K-satisfiability (K-SAT) problem", The 3rd KIAS Conference on Statistical Physics, KIAS, Seoul, Korea, July 2008.
- "Googling social interactions: Web search engine based social network construction", Physics of distributed information systems (PhysDIS), NORDITA, Stockholm, Sweden, May 2008.
- "Bipartite network analysis on protein complexes in *Saccharomyces cerevisiae*", APPC 10: The 10th Asia Pacific Physics Conference, POSTECH, Pohang, Korea, August 2007.
- "Social network analysis based on WWW search engine", 23rd International Conference on Statistical Physics of the International Union of Pure and Applied Physics (STATPHYS 23), Genoa, Italy, July 2007.
- "Social network analysis based on WWW search engine", American Physical Society March Meeting, Denver, Colorado, the United States, March 2007.
- "Statistical properties of sampled networks", Dynamics Days Asia Pacific 4: the 4th International Conference on Nonlinear Science, POSTECH, Pohang, Korea, July 2006.
- "Effects of substrate network topologies on competition dynamics", Dynamics Days Asia Pacific 4: the 4th International Conference on Nonlinear Science, POSTECH, Pohang, Korea, July 2006.
- "Random field Ising model on networks with heterogeneous connections", The 2nd KIAS Conference on Statistical Physics, KIAS, Seoul, Korea, July 2006.
- "Statistical Properties of Sampled Networks", American Physical Society March Meeting, Baltimore, Maryland, the United States, March 2006.

Additional Service

Workshop/Conference Organizing

- Organizer, The 23rd KIAS-APCTP Winter School on Statistical Physics, High1 Resort, Jeongseon, Korea, January 5–9, 2026.
- Organizer, The 23rd Workshop for Statistical Physics 2025, Pukyong National University, Busan, Korea, August 18–20, 2025.
- Organizer, Power-net 2023 (NetSci 2023 Satellite Workshop), Vienna, Austria, July 10, 2023.
- Organizer, Modeling on Ecology and Evolution Workshop (MEEW), Seoul, Korea, October 27–29, 2022.

- Organizer, Power-net 2022 (NetSci 2022 Satellite Workshop), online, July 12, 2022.
- Organizer, Power-net 2021 (Networks 2021 Satellite Workshop), online, July 1, 2021.
- Publicity/Social Media Chair, NetSci-X 2020: International School and Conference on Network Science, Tokyo, Japan, January 20–23, 2020.
- Social Media Chair, NetSci2016: International School and Conference on Network Science, Seoul, Korea, May 30–June 3, 2016.
- Organizer, When Complex Networks Meet Complex Data: Higher-Order Models in Network Science (NetSci2016 Satellite Workshop), Seoul, Korea, May 30, 2016.
- Organizer, Workshop on Time-Dependent and Multiplex Networks, University of Oxford, Oxford, United Kingdom, July 8–9, 2013.

Journal Reviewer

Nature Communications, Nature Human Behaviour, Physical Review Letters, Physical Review A, Physical Review Research, APS Open Science, Physical Review A, Chaos: An Interdisciplinary Journal of Nonlinear Science, Physica A, EPL (Europhysics Letters), The European Physical Journal–Special Topics, The European Physical Journal B, Journal of Statistical Mechanics: Theory and Experiment (JSTAT), Journal of Statistical Physics, Journal of the Korean Physical Society, New Physics: Sae Mulli (새물리), Scientific Reports, PLOS ONE, The IMA Journal of Applied Mathematics, European Journal of Applied Mathematics, Journal of Complex Networks, Network Science, Applied Network Science, Social Network Analysis and Mining, Palgrave Communications, IEEE Transactions on Network Science and Engineering, Complexity, Entropy, Habitat International, IET Systems Biology, Science China Information Sciences.

Conference/Workshop Program Committee

- NetSci2027: International School and Conference on Network Science, Dresden, Germany, May 17–21, 2027.
- NetSciX 2026: International School and Conference on Network Science, Auckland, New Zealand, February 17–20, 2026.
- NetSci2025: International School and Conference on Network Science, Maastricht, the Netherlands, June 2–6, 2025.
- NetSci2024: International School and Conference on Network Science, Québec City, Canada, June 16–21, 2024.
- NetSciX 2024: International School and Conference on Network Science, Venice, Italy, January 22–25, 2024.
- NetSci2023: International School and Conference on Network Science, Vienna, Austria, July 10–14, 2023.
- NetSciX 2023: International School and Conference on Network Science, Buenos Aires, Argentina, February 7–10, 2023.
- NetSci2022: International School and Conference on Network Science, online, July 25–29, 2022.
- NetSciX 2022: International School and Conference on Network Science, Porto, Portugal, February 8–11, 2022.
- IC^2S^2: 7th Annual International Conference on Computational Social Science, online and ETH Zürich, Zürich, Switzerland, July 27–31, 2021.
- NetSci2020: International School and Conference on Network Science, online, September 17–25, 2020.
- WebSci2020: The 12th International ACM Conference on Web Science 2020, online, July 6–10, 2020.
- IC^2S^2: 6th Annual International Conference on Computational Social Science, online, July 17–20, 2020.
- NetSci-X 2020: the International School and Conference on Network Science, Tokyo, Japan, January 20–23, 2020.
- IC^2S^2: 5th Annual International Conference on Computational Social Science, Amsterdam, the Netherlands, July 17–20, 2019.

- Conference on Complex Systems (CCS) 2019, Nanyang Technological University, Singapore, September 30–October 4, 2019.
- NetSci2019: International School and Conference on Network Science, Burlington, Vermont, the United States, May 27–31, 2019.
- IC^2S^2: 4th Annual International Conference on Computational Social Science, Evanston, Illinois, the United States, July 12–15, 2018.
- Social Informatics 2018 (Socinfo 2018), Saint Petersburg, Russia, September 25–28, 2018.
- Social Informatics 2017 (Socinfo 2017), Oxford, United Kingdom, September 13–15, 2017.
- Complex Networks 2017: The 6th International Conference on Complex Networks and their Applications, Lyon, France, November 29–December 1, 2017.
- NetSci2017: International School and Conference on Network Science, Indiapolis, Indiana, the United States, June 19–23, 2017.
- Complex Networks 2016: The 5th International Workshop on Complex Networks and their Applications, Milan, Italy, November 30–December 2, 2016.
- The 8th International Conference on Social Informatics (SocInfo 2016), Seattle, Washington, the United States, November 11–14, 2016.
- 2016 International Conference on Computational Social Science (ICCSS), Evanston, Illinois, the United States, June 23–26, 2016.
- International Conference on Computational Social Science (ICCSS), Helsinki, Finland, June 8–11, 2015.
- The 9th International AAAI Conference on Web and Social Media (ICWSM-15), Oxford, United Kingdom, May 26–29, 2015.
- Computational Social Science (CSS) 2014 [European Conference on Complex Systems 2014 (ECCS'14) Satellite Workshop], Lucca, Italy, September 24–25, 2014.

Outreach on Network Science for School Students in United Kingdom

- Places and Dates: Arden School, Solihull (November 19, 2012), Somerville College, University of Oxford (November 24, 2012), The Discovery Academy, Stoke-On-Trent (December 3, 2012), Somerville College, University of Oxford (May 11, 2013: YouTube video below), Brentford School for Girls, Brentford (November 20, 2013), Ormiston Sir Stanley Matthews Academy, Stoke-On-Trent (February 3, 2014).
- Video Clip: Sharing the beauty of networks. `http://www.youtube.com/watch?v=9dcdjcyA-8E`, September 4, 2013.
- Related article: H. A. Harrington, M. Beguerisse Díaz, M. P. Rombach, L. M. Keating, and M. A. Porter, "Commentary: Teach network science to teenagers", Network Sciences **1**, 226 (2013).

Translation

- "페이스북의 친구추천이 정교해진 이유" ("The reason why the friend recommendation on facebook has got accurate" in English) [the interview with László Barabási during NetSci 2016], Tech M, June 9, 2016. `http://techm.kr/bbs/board.php?bo_table=article&wr_id=2139`.
- The Korean version of Network Literacy Project (in collaboration with Mi Jin Lee): "네트워크 이해하기 (in Korean)". `https://sites.google.com/site/lshlj82/networkliteracy`, April 27, 2015.

Background and Skills

programming and data analyses with C, Python, MATLAB, R, etc.; UNIX/LINUX/Mac OS systems; large-scale network analysis tools (Pajek, NetworkX package of Python, etc.); agent-based modeling; Web data mining; statistical physics; Monte Carlo simulation; finite-size scaling analysis; video editing with Final Cut Pro.

Last updated: November 5, 2026